

Zahra Golpayegani

ML Engineer
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<https://zahragolpa.github.io/>

About Me

I am a machine learning engineer with **4+ years of experience scaling and optimizing** large language models for **efficient training and inference**. My work focuses on **hardware-aware** performance optimization, **parallel and distributed workloads**, and **end-to-end ML infrastructure** across training and inference stacks.

Experience

Huawei Canada – *Researcher*

Sep 2024 – Present

- Integrated **async RL training with multi-turn tool-call support for Ascend NPUs** in **VeRL** (CANN-recipes-train repository), enhancing long-horizon reasoning performance and extending **evaluation** coverage with benchmarks such as LiveCodeBench ([PR #101](#), [PR #102](#)).
- Implemented a **block-wise speculative decoding method** (proposed by Google) and integrated it into **vLLM and vLLM-Ascend**, improving inference **throughput by up to 10.2%** and reducing **TPOT by 9.2%** ([PR #40819](#), [PR #7808](#)).
- Co-developed a simulation library for **distributed LLM training and serving workloads**, incorporating kernel-level FLOP and memory-based **performance modeling** for new LLM architectures, enabling **systematic throughput and latency analysis** across hardware and serving configurations.
- Implemented an **Attention-FFN Disaggregation (AFD)** simulation framework for SOTA LLMs (e.g., **DeepSeek NSA, DeepSeek DSA, TurboS**) to evaluate serving efficiency in disaggregated pipelines.
- Identified **performance bottlenecks** in LLM workloads on **Huawei Ascend accelerators** through systematic **profiling**, and proposed targeted **optimizations** to improve compute efficiency.
- Benchmarked parallelization strategies—including **tensor, expert, context, sequence, and data parallelism**—for models such as **DeepSeek-NSA** and **MiniMax-Text-01**, quantifying impacts on memory usage, communication overhead, and throughput.
- Accelerated pre-training of **LLaMA-based models** using conditional computation, reducing training cost by **up to 16%** and **improving MFU by 3%** without significant accuracy loss.
- Proposed a **test-time training** method that extends GPT-Large and Phi-2 **context length by up to 32×** while **improving accuracy by 30%** ([link to paper](#)).

Zetane Systems – *AI Engineer*

Dec 2021 – Aug 2024

- Co-designed and developed [ZetaForge](#), an AI platform automating prompt-to-pipeline generation using LLMs.
- Developed complex end-to-end multi-agent AI pipelines leveraging state-of-the-art libraries such as **LangChain** and **openai-python** ([blog post](#)).
- Implemented a robust **MLOps pipeline** (end-to-end machine learning operations workflow) featuring data and model validation, preprocessing, model training with **automated experiment tracking**, and continuous model improvement ([open-source project](#)).

- Contributed to flagship products including Protector—a platform for assessing vision model reliability in real-world conditions.
- Delivered product demos and technical talks to prospective clients and international audiences at events including Scale AI, World Summit AI, and ALL IN.

XAI Lab, Concordia University – *Research Assistant*

Sep 2021 – Jan 2024

Tools: Python, PyTorch, OpenCV, MATLAB, LaTeX

- Investigated the interplay between **model robustness, accuracy, and shape bias** under Professor Nizar Bouguila’s supervision; published findings at **CRV 2023**.
- Proposed a novel image compression algorithm inspired by **Singular Value Decomposition (SVD)** to optimize robustness of vision models trained on compressed data; work published at **ICPRAM 2024**.

Skills

Programming Languages: Python, C/C++, CUDA	Deep Learning Frameworks: PyTorch, vLLM, VeRL, Megatron-LM, HuggingFace, LLaMA Factory
MLOps & DevOps: Docker, LangChain, MLflow, Git	

Education

Concordia University , MSc Information Systems Engineering	2024
GPA: 4.0 / 4.0	
Thesis: Enhancing Deep Learning Model Robustness: Insights from Out of Distribution Data Augmentation and an Innovative Image Compression Technique	
Supervisor: Prof. Nizar Bouguila	
Amirkabir University of Technology , BSc Computer Science	2020
GPA: 3.7 / 4.0	
Final Project: Food9K: Detecting Food in Social Media Images Using YOLOv3	
Supervisor: Prof. Mohammad Akbari	

Publications

- ETT: Expanding the Long Context Understanding Capability of LLMs at Test-Time. 2025. arXiv. (link)
- PatchSVD: A Non-Uniform SVD-Based Image Compression Algorithm. 2024. ICPRAM. (link)
- Clarifying myths about the relationship between shape bias, accuracy, and robustness. 2023. CRV. (link)

Awards

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| • Campaign Hero Award , Huawei | 2025 |
| Honored for delivering rapid, high-impact results on a key internal campaign at Huawei HQ. | |
| • Future Star Award , Huawei | 2025 |
| Recognized for outstanding performance and potential, highlighting my contributions. | |
| • Best Paper Award Candidate , ICPRAM | 2024 |
| Nominated for research on model compression and robustness. | |
| • Conference and Exposition Award , Concordia University | 2023 |
| Awarded for excellence in presenting graduate-level research. | |
| • Accelerate Explore Award , Mitacs | 2021 |
| For conducting applied AI research in collaboration with industry partners. | |
| • Merit Scholarship Entrance Award , Concordia University | 2021 |
| Awarded for high academic achievement upon admission to MSc program. | |
| • 2nd Place, Worldwide , RoboCup Junior Soccer League | 2013 |
| <u>RoboCup World Competitions</u> – international robotics tournament. | |